Matched degree-83 IC and optimized rho

Actual cold processes, genuine reused setup, and a larger fresh target panel

Exact curve: icv1-f2m83-t6151469093347-9eba6cd5

32 fresh targets; three real shared-setup cohorts; eight actual singleton pairs

Ratios are IC/rho; values below 1 favor IC

A new addendum precedes the preserved earlier 70 pages

Actual IC/rho mean complete-process ratio: 377.435691 for eight cold singleton pairs; 15.615702 for three genuine 32-target cohort pairs. Values above 1 favor rho. The 32-target query comparison and crossover model remain secondary.

Contents

The actual complete-process boundaries lead this report. Query distributions, memory and the additive break-even model are secondary. Source links bind every new value to native records. Historical evidence is unchanged.

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SECTION 01

Actual complete-process comparison

> All 32 solver processes completed and all 218 answers passed independent scalar replay. Actual cold singleton means are IC 1159.349882 s and rho 3.071649 s; actual 32-target cohort means are IC 1232.975217 s and rho 78.957400 s.

The two frozen methods solve the same point-only inputs under one recorded resource envelope. IC is the existing wide koblitz_orbit_dlp_fast producer with the retained K600 factor base. Rho uses Fold field reduction, reference u64 scalar arithmetic, 256 lockstep lanes, DP6, fruitless escape off and step-cap factor 2000. This study performs no further configuration selection on the new points.

Actual measured workload	IC wall (s)	Rho wall (s)	IC/rho and native interval
Eight fresh singleton pairs	1159.349882	3.071649	377.435691 / [311.812847, 499.010210]
Three real 32-target cohort pairs	1232.975217	78.957400	15.615702 / [14.743543, 17.614848]

Cold here means an actual fresh process given the already materialized K600 artifact. Historical factor-base selection and construction cost is unavailable and remains an additive gap for a curve-only cold-start comparison. The shared workload initializes each arm once for 32 points; every target receives fresh walk, distinguished-point and collision state.

All 32 processes completed: 208 main matched answers and 10 A/A answers. Independent native replay passes for all 218 scalars, all 104 IC four-point witnesses and all five same-work A/A pairs, with zero invalid or missing answers. The reservation records 2,654 samples and zero contended samples. Wall times remain conditional on one guest host; calibrated S, ecbench L2 and independent-host admission remain unset.

Sources: Protocol, Native statistics, Independent replay, Collector seal.

SECTION 02

Curve, bases and invariants held fixed

The retained model is icv1-f2m83-t6151469093347-9eba6cd5: $y^2+xy=x^3+x^2+1$ over GF(2)[z]/(z^83+z^7+z^4+z²+1), prime subgroup r=8569786107849059, cofactor 1128547018, and G=(661424465139818486029201 5,886826041002924939153952 9). The field degree is 83; the subgroup has about 52.928 bits. The signed Frobenius quotient uses 166 phases and eigenvalue λ=2998004823950457.

Representation or invariant	Role in the measured algorithm	Identity / accounting status
Binary field and polynomial basis	Exact field arithmetic and curve representation	Tail 149; no tail-45 or a=0 substitution
Normal-basis conversion and signed Frobenius quotient	Shared coordinate/quotient machinery; 166 signed phases	Charged once per genuine process
K600 factor-base point set	600 columns and 99,600 signed orbit points for IC	Artifact parsing, index and rank/log reconstruction charged; original selection unavailable
Reference scalar arithmetic and rho policy	Fold field, u64 scalars, 256 lockstep lanes, DP6, escape off	Frozen prior selected configuration; no new tuning

These are distinct objects: a field representation, an orbit quotient, an IC factor-base point set and a solver policy. They can share exact curve and subgroup invariants without sharing a cost boundary. This follow-up changes measurement coverage and workload accounting; it establishes no new generic counting exponent or algebraic invariant.

Registry representation EC1N83Ce1h35c7ced6b497; curve UID urn:ec-record:1:sha256:35c7ced6b497231fbc82fe24db596fd55379a87356962dad47a7dc774388ce9d. K600 file SHA-256 0d290b14a350db6271c9a49f0cd6ba246edd8c9bc6754387ad1ceebf08ec0c09; declared point-set digest d4a4da26d6e7cb09367de922d9806631fe49989f70d378cae76f347f8d76a9f7. Exact field, subgroup, generator and point-set identities are retained rather than inferred from degree alone.

Sources: Exact curve registry, Frozen protocol, Build/source pins.

SECTION 03

Fresh panel and frozen execution order

Native C++ generation uses unbiased nonzero scalar rejection sampling with SplitMix64 seed 202610146301. It excludes the signed orbits of the K600 representatives, the original exposed target, all 16 prior optimization targets and each accepted new target. There are 617 initially excluded orbits. Solvers receive point-only files; known validation scalars remain in a checker-only sidecar.

Prospective component	Processes / answers	Order and purpose
Five identical rho A/A pairs	10 processes / 10 answers	target_01, equal seed inside each pair; noise diagnostic
Three genuine shared cohorts	6 processes / 192 matched answers	32 points per arm, target orders rotated by 0/11/22; second rotated order reversed
Eight actual singleton cold pairs	16 processes / 16 matched answers	target_01 through target_08; predeclared alternating arm order
Total prospective matrix	32 processes / 218 answers	208 main matched answers plus 10 A/A answers; every attempt retained

All 32 fresh points and both older eight-point streams pass independent native regeneration, scalar/subgroup checks and signed-orbit exclusions. Shared rho seeds are 202610146501–596, assigned by frozen target index and cohort; actual cold seeds are 202610146601–608; A/A seeds are 202610146411–415. IC rank seed stays 29. Every actual target order and point-file/seed-file digest is in the frozen matrix.

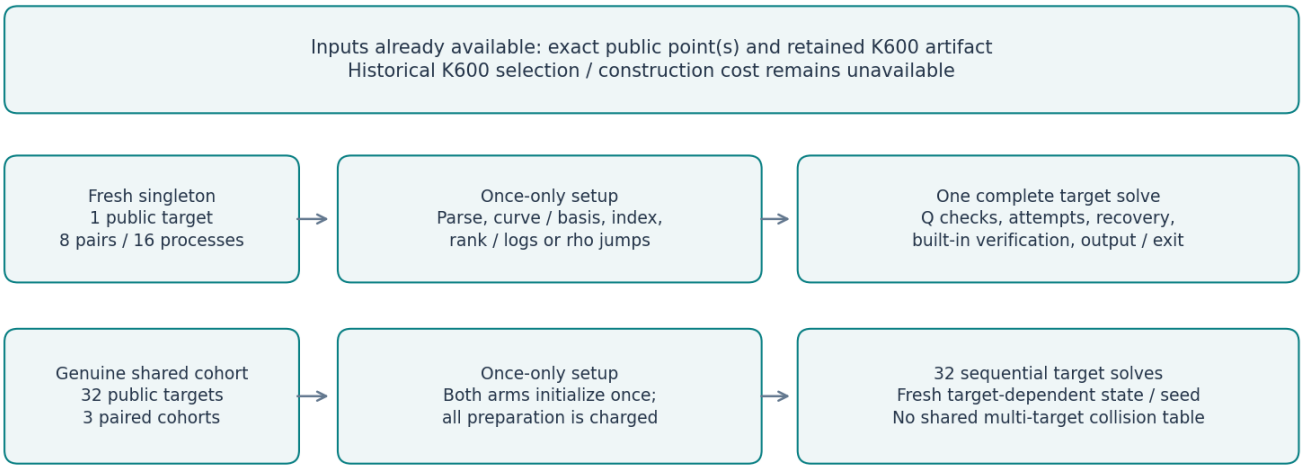
No fastest-seed selection, target deletion, adaptive pruning or policy change is permitted. Sampling is conditional on orbit exclusions and without replacement. Target and preparation repetitions are distinct: the online panel has 32 target blocks sharing three IC preparations.

Sources: Fixture generator, Independent fixtures, Frozen execution matrix.

SECTION 04

What the clocks include

Measured process boundaries



Producer query timers are secondary: IC and rho Q-validation boundaries differ.

Outside both clocks: target generation, research / tuning, compilation, independent post-hoc audit.
Cold denotes a fresh process, not an OS-cache flush. Complete process costs lead this report.

Actual fresh singleton and genuine reused-setup process boundaries. Public-point generation, compilation and external research audit are outside both arm process clocks. Historical K600 selection/construction cost is unavailable. Run run-20261014T1940Z.

Actual process wall includes launch, public input parsing, curve/basis setup, IC index and rank/log reconstruction or rho jumps, Q checks, solving, scalar recovery, built-in verification, output and exit. A whole-process timeout and native wait4 meters apply equally to both arms.

Stock IC target_ms omits input/Q-validation/output intervals that rho online includes. Those producer query timings stay secondary and retain the mismatch. The five exported IC stage fields and their residual are shown; they are not silently redefined as five exhaustive mutually exclusive end-to-end phases.

Producer preparation includes IC index/rank/log work and rho curve/normal-basis/seed-specific jumps. Independent final V3 scalar/witness/file replay took 1.167454 s wall, 1.089206 s user CPU and 0.075961 s system CPU, peak RSS 109,332 KiB, separately metered without observed contention. That is one post-hoc research-audit pass over the campaign, not a production per-target verification estimate. Built-in solver verification is already inside each process cost.

Sources: IC timer source, Rho timer source, Audit meter.

SECTION 05

Recorded phase accounting and residuals

All 104 main IC answers retain the five producer stage fields: 96 queries in shared cohorts and eight singleton queries. The native supplement computes the signed target-clock-minus-stage-sum residual. These fields describe the producer clock; they do not repair its omitted Q-validation/input/output boundary.

Native IC field (ms)	Mean	Median	p95	Maximum
Query-origin preparation	0.000026	0.000030	0.000030	0.000040
Point decomposition excluding relation checks	1718.020904	827.241681	6604.401760	9481.881343
Relation checks	0.014410	0.012750	0.024960	0.108736
Scalar descent / recovery	0.001578	0.000726	0.001130	0.026090
Recovery group check	0.136032	0.120762	0.163148	1.243078
Target clock minus five-phase sum	0.000921	0.000781	0.001069	0.013611

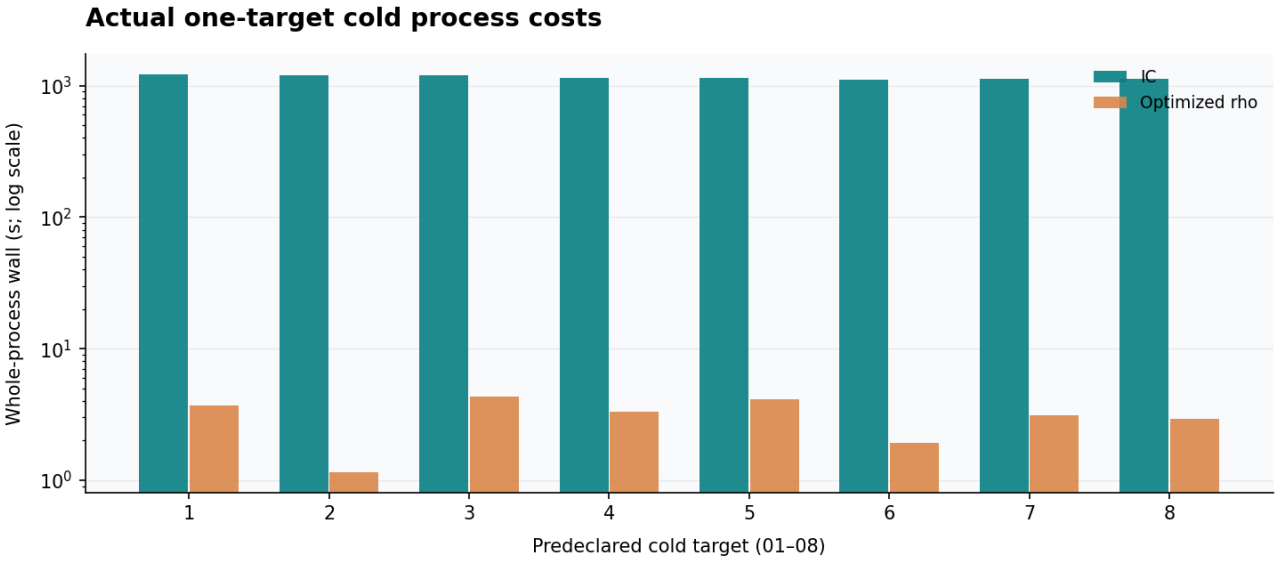
Native IC residual (target_ms minus five_phase_sum_ms), n=104: mean 0.000921423 ms, median 0.000781000 ms, minimum 0.000300000 ms and maximum 0.013611000 ms, with zero negative rows. This small residual is retained; the larger omitted input/Q-validation/output boundary remains a separate issue. The supplement's source-file hashes and JSON/JSONL pointers bind each recorded field.

Rho retains exact nanosecond phase and residual counters in its own raw target records, with A/A cases identified separately. Process-level supervisor-minus-producer residuals, CPU and RSS remain in the native supplement with source-file hashes and JSON/JSONL pointers. No phase sum or residual is recalculated by the PDF renderer.

Sources: Native signed phase/residual supplement, Native phase exporter source, Raw producer fields.

SECTION 06

Eight actual cold singleton pairs



Primary complete-process boundary; both arms independently verified
Eight actual paired public targets · retained K600 input
run-20261014T1940Z · records/postprocess-20261015T0018Z/statistics.json

All eight predeclared actual singleton process pairs, seconds on a logarithmic y-axis. Primary complete-process boundary, given retained K600; every original pair independently verifies. Run run-20261014T1940Z.

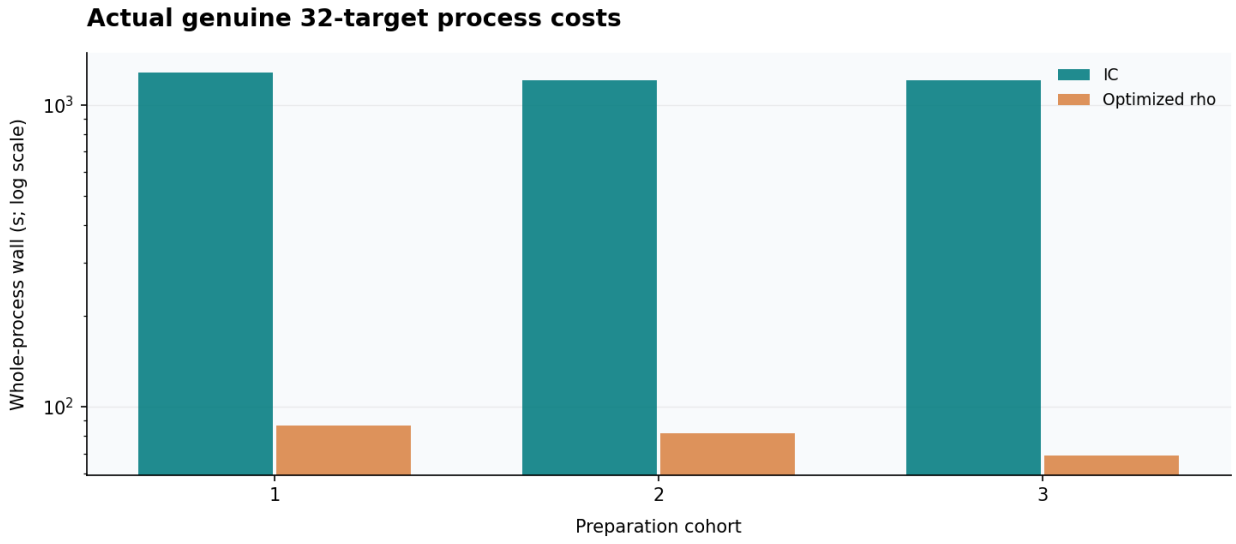
Target	IC process (s)	Rho process (s)	IC/rho / status
target_01	1222.138607	3.720382	328.498148 / verified
target_02	1203.297036	1.143097	1052.663582 / verified
target_03	1193.387645	4.327487	275.769196 / verified
target_04	1146.736518	3.315835	345.836421 / verified
target_05	1140.785160	4.117988	277.024876 / verified
target_06	1109.371728	1.915267	579.225622 / verified
target_07	1131.835702	3.121473	362.596633 / verified
target_08	1127.246662	2.911660	387.149111 / verified

Native paired eight-target percentile 95% interval for the mean-cost ratio: [311.812847, 499.010210]. IC process median 1143.760839 s (min 1109.371728, max 1222.138607); rho median 3.218654 s (min 1.143097, max 4.327487). All eight original point pairs are included; given retained K600, without OS-cache flushing.

Sources: All cold rows and native estimator, Raw cold process K01-ic.

SECTION 07

Three genuine reused-setup cohorts



Primary: actual shared infrastructure and every target solve charged
Three paired preparation cohorts · 32 points per arm/process
run-20261014T1940Z · records/postprocess-20261015T0018Z/statistics.json

All three actual 32-target process pairs, seconds on a logarithmic y-axis. Each arm initializes infrastructure once and charges every target. Rho target-dependent collision state is reset. Only three preparation clusters. Run run-20261014T1940Z.

Cohort	IC process / setup (s)	Rho process / setup (s)	Complete status
1	1282.605726 / 1222.727201	86.376504 / 0.254761	Both verified; all 32 points
2	1204.820756 / 1149.988148	81.718538 / 0.271938	Both verified; all 32 points
3	1211.499169 / 1156.448203	68.777158 / 0.241819	Both verified; all 32 points

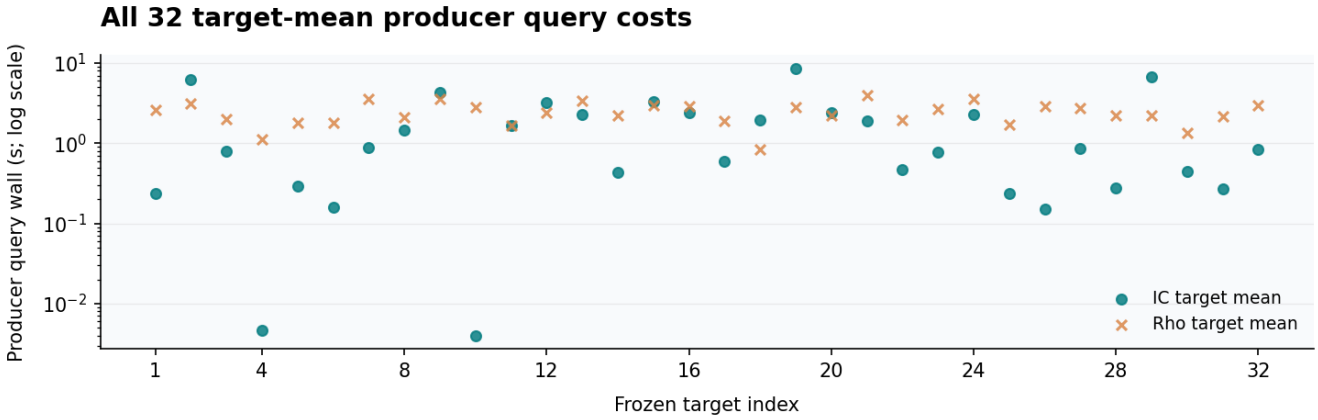
The primary shared-workload estimate uses mean actual process cost over three paired cohorts, not a synthetic sum of singleton launches divided by 32. Its native paired bootstrap conditions on this one host and only three preparation clusters; it does not describe independent-host uncertainty or robust extreme tails.

Native mean actual process ratio IC/rho 15.615702, paired-cohort 95% interval [14.743543, 17.614848]. Each arm has exactly three observed whole-process costs. No process timeout, failure or target omission is converted into a zero-cost observation.

Sources: All shared cohorts and native estimator, Raw cohort C01-ic.

SECTION 08

Producer query distributions and slow cases



Secondary query comparison; not an end-to-end speedup
32 paired target blocks · three shared preparations · unequal validation boundaries
run-20261014T1940Z · records/postprocess-20261015T0018Z/statistics.json

All 32 native target means in seconds on a logarithmic y-axis, three query observations per arm/target. The IC and rho producer validation/input/output boundaries differ; this is a secondary diagnostic. Run run-20261014T1940Z.

Native descriptive statistic	IC	Rho	Scope
Mean query (s)	1.760364	2.457289	32 target means, three repeats each
Median query (s)	0.852103	2.332984	32 target means, three repeats each
p90 query (s)	4.182302	3.523073	32 target means, three repeats each
p95 query (s)	6.412727	3.579652	32 target means, three repeats each
Maximum query (s)	8.590054	4.016571	32 target means, three repeats each
Mean query (s)	1.760364	2.457289	96 draws sharing three preparations
Median query (s)	0.833230	2.168825	96 draws sharing three preparations
p90 query (s)	4.401029	4.401391	96 draws sharing three preparations
p95 query (s)	6.677992	5.016997	96 draws sharing three preparations
Maximum query (s)	9.482010	6.193312	96 draws sharing three preparations

Secondary ratio of arithmetic mean target costs IC/rho 0.716385, 32-target-block 95% interval [0.460389, 1.013127]. Crossed target/cohort sensitivity interval [0.452168, 1.046078]. Descriptive geometric mean of paired target-mean ratios 0.315622. These clocks have unequal validation boundaries and do not establish the complete-process winner.

Means, medians and tail quantiles answer different questions. Conventional even-sample medians use midpoint interpolation in native analysis. Every slow or regressing point remains in the panel; successful subset summaries never turn a failed or missing solve into a win.

Sources: All native query distributions/intervals, Native statistical source.

SECTION 09

Complete query rows: targets 01–16

Each row contains the native three IC producer query observations, three rho observations and native target means/ratio. Units and unequal Q-validation boundaries remain explicit. Seeds and full point hashes are in the source records.

Target	IC queries (s)	Rho queries (s)	Native means (s) / IC/rho
target_01	0.234867, 0.260464, 0.222896	1.640854, 5.196469, 0.989932	0.239409 / 2.609085 ratio 0.091760
target_02	6.642331, 5.957789, 5.849844	3.296084, 3.590678, 2.502273	6.149988 / 3.129678 ratio 1.965055
target_03	0.909376, 0.777519, 0.723368	1.429782, 2.474638, 2.034428	0.803421 / 1.979616 ratio 0.405847
target_04	0.004565, 0.004735, 0.004703	1.616102, 0.872278, 0.917898	0.004667 / 1.135426 ratio 0.004111
target_05	0.311761, 0.297816, 0.265460	3.087775, 0.600333, 1.784729	0.291679 / 1.824279 ratio 0.159887
target_06	0.187674, 0.146220, 0.146495	2.061500, 1.040531, 2.277989	0.160130 / 1.793340 ratio 0.089291
target_07	0.988067, 0.827510, 0.815178	6.193312, 0.657449, 3.775011	0.876918 / 3.541924 ratio 0.247582
target_08	1.619075, 1.344187, 1.362457	1.579708, 3.997688, 0.814403	1.441906 / 2.130600 ratio 0.676761
target_09	4.568501, 4.233557, 4.050278	3.845046, 1.977733, 4.778620	4.284112 / 3.533800 ratio 1.212324
target_10	0.004085, 0.003979, 0.003934	0.847963, 4.154483, 3.570212	0.003999 / 2.857553 ratio 0.001400
target_11	1.635520, 1.628618, 1.670129	2.819685, 1.479100, 0.699827	1.644756 / 1.666204 ratio 0.987128
target_12	3.164274, 3.178024, 3.194376	2.177541, 2.973375, 2.087461	3.178891 / 2.412792 ratio 1.317515
target_13	2.291834, 2.260263, 2.302585	3.456101, 4.663401, 2.160109	2.284894 / 3.426537 ratio 0.666823
target_14	0.448802, 0.421863, 0.427136	0.777461, 5.015016, 0.843483	0.432600 / 2.211987 ratio 0.195571
target_15	3.381695, 3.253157, 3.163199	2.830017, 3.089726, 3.080597	3.266017 / 3.000113 ratio 1.088631
target_16	2.575803, 2.375660, 2.342953	2.228766, 4.511234, 2.063196	2.431472 / 2.934399 ratio 0.828610

Sources: All native target rows, Raw query input.

SECTION 10

Complete query rows: targets 17–32

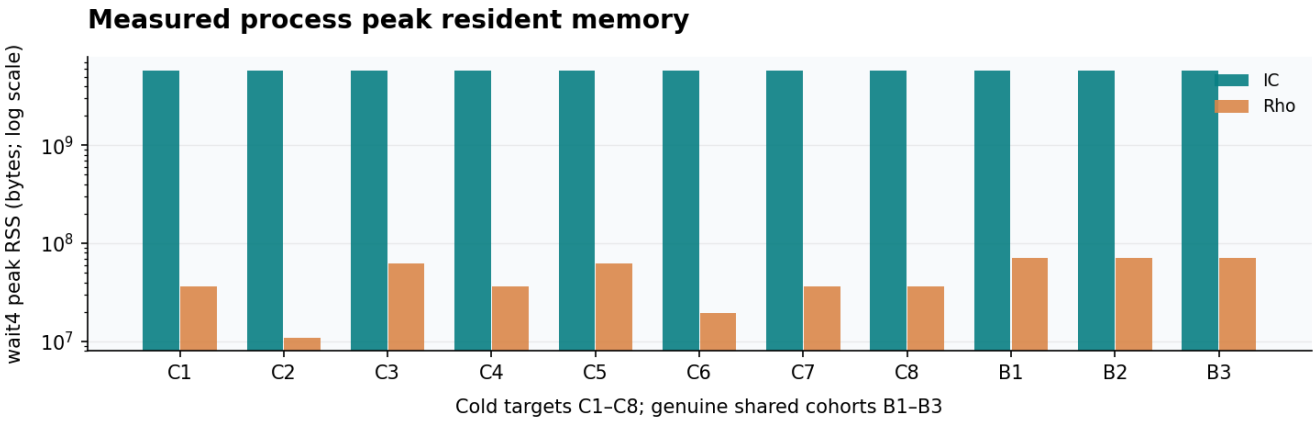
The same native accounting and formatting apply to every point. These rows are retained rather than replacing the panel with a pooled median or selected fastest cases.

Target	IC queries (s)	Rho queries (s)	Native means (s) / IC/rho
target_17	0.620815, 0.592335, 0.589533	0.262199, 2.019367, 3.439275	0.600894 / 1.906947 ratio 0.315108
target_18	2.187152, 1.832016, 1.841578	0.963085, 0.730592, 0.826744	1.953582 / 0.840140 ratio 2.325304
target_19	9.482010, 8.086817, 8.201336	5.500752, 1.531207, 1.373199	8.590054 / 2.801719 ratio 3.065994
target_20	2.596347, 2.399435, 2.301716	2.213324, 3.360088, 1.135344	2.432500 / 2.236252 ratio 1.087757
target_21	2.045828, 1.783484, 1.906819	4.287396, 2.836442, 4.925875	1.912044 / 4.016571 ratio 0.476039
target_22	0.484078, 0.457909, 0.483056	3.257985, 0.975511, 1.693946	0.475014 / 1.975814 ratio 0.240415
target_23	0.784440, 0.770213, 0.794702	4.291549, 0.912472, 2.925784	0.783118 / 2.709935 ratio 0.288980
target_24	2.349436, 2.251366, 2.202186	2.758987, 5.241807, 2.876498	2.267663 / 3.625764 ratio 0.625430
target_25	0.255149, 0.216389, 0.238129	2.495483, 1.753540, 0.910344	0.236556 / 1.719789 ratio 0.137549
target_26	0.139901, 0.168311, 0.149233	4.233758, 2.779285, 1.746992	0.152481 / 2.920012 ratio 0.052219
target_27	0.857479, 0.813129, 0.910952	1.330264, 5.022941, 1.790285	0.860520 / 2.714497 ratio 0.317009
target_28	0.269494, 0.262076, 0.298113	4.057517, 1.126848, 1.575164	0.276561 / 2.253176 ratio 0.122743
target_29	7.026236, 6.390347, 6.784973	1.931769, 3.711723, 1.042105	6.733852 / 2.228532 ratio 3.021653
target_30	0.434742, 0.476855, 0.429565	1.754524, 0.376710, 1.863671	0.447054 / 1.331635 ratio 0.335718
target_31	0.267509, 0.280338, 0.265773	2.712757, 1.046975, 2.798944	0.271207 / 2.186225 ratio 0.124052
target_32	0.827256, 0.838951, 0.864852	4.111872, 1.660656, 3.164147	0.843686 / 2.978892 ratio 0.283222

Sources: All native target rows, Raw query input.

SECTION 11

Memory, CPU and retained failures



Observed RSS, not RLIMIT_AS allocation and not additive across sequential processes
Eight cold pairs and three shared pairs · process peaks shown separately
run-20261014T1940Z · records/postprocess-20261015T0018Z/statistics.json

Native wait4 peak RSS in bytes for all eight cold pairs and three genuine shared pairs. These sequential process peaks are not additive concurrent footprints. Run run-20261014T1940Z.

Workload / arm	CPU / peak RSS	Complete / failure status
IC / 8 cold	CPU mean 1159.264837 s RSS min / max 5811519488 / 5811724288 bytes	All original processes verified; costs retained
RHO / 8 cold	CPU mean 3.063495 s RSS min / max 10928128 / 62115840 bytes	All original processes verified; costs retained
IC / 3 shared	CPU mean 1232.883520 s RSS min / max 5811548160 / 5811691520 bytes	All original processes verified; costs retained
RHO / 3 shared	CPU mean 78.941576 s RSS min / max 70713344 / 70823936 bytes	All original processes verified; costs retained

Each run retains exit/signal status, timeout flags, the raw memory-limit indicator, scalar-validation status and internal rho capped-walk counters. A null memory-limit indicator remains unavailable. Internal rho lane restarts are charged algorithmic work; zero whole-process timeouts does not imply zero internal restarts.

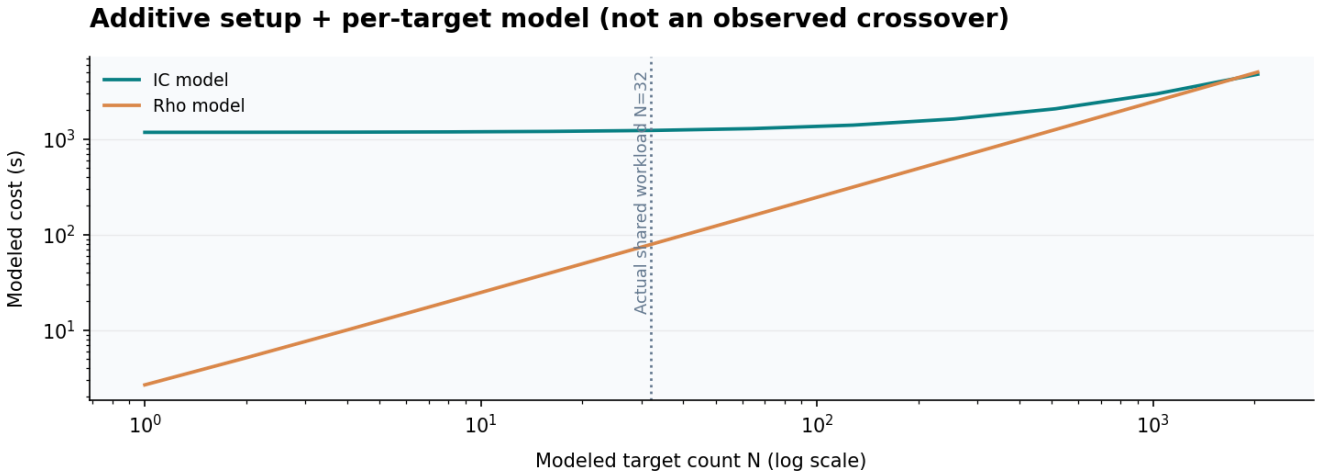
All 32 processes completed under the 12-GiB RLIMIT_AS envelope with no whole-process timeout, reported solve failure or invalid scalar answer. The memory_limit_hit indicator is null for every process; memory-limit-hit detection is unavailable and is not inferred from exit status. Per-target rho work counters below retain internal capped-walk restarts. IC peak RSS is about 5.81 billion bytes on every process; exact native wait4 values remain in the records. No consumed work is dropped from the complete-process estimators.

Native rho work counters: the 104 main answers retain 1001130567 walk steps and 4488 internal capped-walk restarts; the 10 A/A answers retain 558 restarts. All 114 rho answers retain 5046 restarts. These algorithmic events and their work remain charged. They are distinct from whole-process timeout or global-cap failure.

Sources: Exact resource/work supplement, Native process records.

SECTION 12

Reusable-setup break-even is a model



MODEL ONLY: N=1,685 point threshold; nonfinite bootstrap fraction 0.0407; unequal query clocks
Native model · target count above 32 is unexecuted extrapolation
run-20261014T1940Z · records/postprocess-20261015T0018Z/statistics.json

Native additive setup-plus-linear-query model. N=1,685 is a modeled point threshold; only N=1 and N=32 complete processes were executed. The nonfinite bootstrap fraction is 0.0407. Query clocks differ and process residual/original K600 construction costs are absent. Run run-20261014T1940Z.

Model field	Native value / status
Native additive setup coefficients	IC 1176.387851 s; rho invariant setup 0.216993 s
Native per-target coefficients	IC query 1.760364 s; rho query 2.457289 s + seed-specific jumps 0.001224 s = 2.458513 s
Point-estimate first favorable target count	N=1,685; modeled and outside executed N=32
Bootstrap finite / nonfinite draws	19,186 / 814 of 20,000; nonfinite fraction 0.0407 retained
Unconditional extended 95% interval	[844, nonfinite / no eventual IC advantage]
Observed complete-process sizes	Actual N=1 and N=32; no larger measured crossover

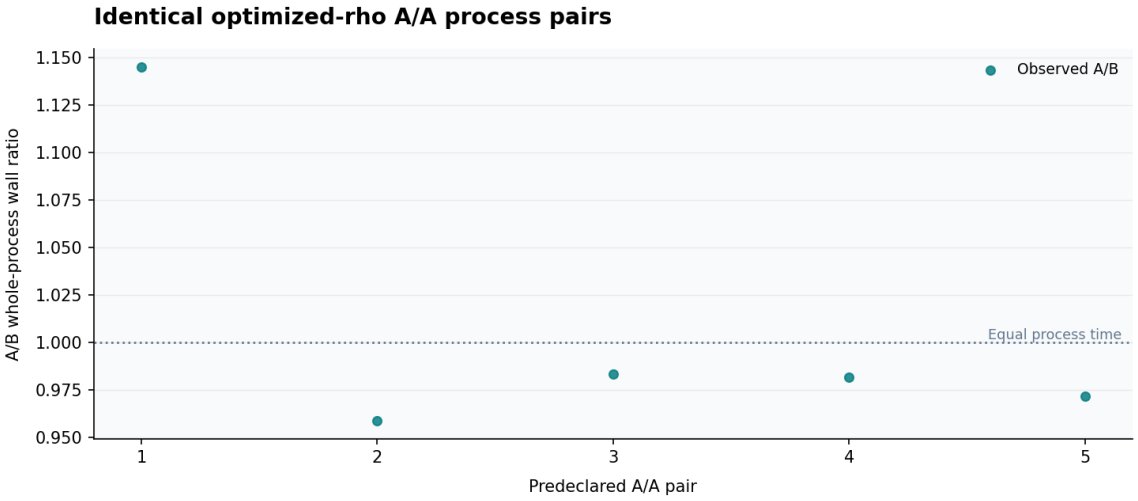
The model assumes additive reusable setup and a stable mean per-target cost. Producer query intervals have unequal Q-validation boundaries, and the original K600 construction cost is absent. Thresholds conditional on those quantities cannot establish a complete production or curve-only crossover. If IC's mean query cost is not lower, the native point estimate has no finite threshold.

Native model point threshold: N=1,685, beyond the only measured shared size N=32. Of 20,000 crossed-bootstrap draws, 814 (fraction 0.0407) have no eventual IC advantage in the linear model. The unconditional extended 95% interval starts at 844 targets and has a nonfinite upper endpoint. Finite-only descriptive quantiles are median 1,602, p025 840 and p975 11,270.125; these condition on finite draws and are not an unconditional confidence interval. Native setup-plus-query prefixes are derived producer-accounted prefixes, not observed prefix wall times. No larger crossover was executed. Projected costs may exceed the current 2100 s cap; larger N requires a new prospective equal-budget protocol.

Sources: Native additive model and finite/nonfinite mass, Frozen native model source.

SECTION 13

A/A noise, isolation and admission



Descriptive noise diagnostic; no confidence band and no correction
Five same-point, same-seed pairs · independent semantic gate passes
run-20261014T1940Z · records/postprocess-20261015T0018Z/statistics.json

Five identical rho A/A singleton process pairs: native A/B ratios, with equality at one. This initial local range is descriptive, not a confidence band or bound on the later multi-hour matrix. Run run-20261014T1940Z.

Envelope / admission	Recorded status
CPU and threads	CPU3; one OS thread per child. The 256 lockstep rho lanes are not OS threads.
Common limits	12 GiB RLIMIT_AS; 2100 s process timeout; native wait4 wall/user/system/RSS. memory_limit_hit is null: detection unavailable, not inferred from exit.
Guest host	AMD EPYC 9V74; x86-64; five visible logical CPUs, cgroup four-CPU/16-GiB envelope; Linux 6.18.44 / KVM
Reservation and contention	Native exclusive guest reservation: 2,654 samples, zero contended samples; physical-host exclusivity unknown
Compiler and profile	Rust 1.99.0 b940084d7eb6a299eb4bfeb8e34901bc051e7ac4; LLVM 23.1.1; both workers use the same release compiler/profile
Unadmitted scope	Calibrated operation S, ecbench L2, physical-host exclusivity and independent-host timing reproduction remain unset

A/A process A/B ratios: 1.145033, 0.958760, 0.983577, 0.981760, 0.971640. Native median 0.981760, minimum 0.958760, maximum 1.145033. Five pairs use identical points, seeds, policies and exported semantic work, independently checked. No noise correction is subtracted from observed method costs.

A guest reservation monitors movable processes and local pressure. Physical-host neighbours and frequency are outside its reach. All timing statements are conditional on the measured host, declared target exclusions, retained materialized base and workload boundaries.

Sources: Native reservation, Native A/A statistics, Build receipt.

SECTION 14

Verification and requirement status

Obligation	Actual status / evidence / remaining gap
Frozen larger matched panel	Verified complete: 32 fresh points, 617 initial orbit exclusions, prospective 32-process matrix; all attempts retained
Actual cold and genuine reused setup	Verified complete: eight singleton pairs and three real 32-target cohort pairs; 208 main answers
Independent scalars / IC witnesses / same-work A/A	PASS: 218 scalars, 104 IC four-point witnesses, five A/A semantic pairs; zero invalid/missing answers
Full IC rank/log transcript	Not exported by the existing wide producer; remains unvalidated
Pre-timing native checks	4,089 library tests passed, zero failed, 117 ignored; four strict-worker, nine controller, 14 frozen-analysis and seven metadata tests passed
Source-linked PDFs and preserved prior report	New source-linked addendum and comprehensive PDF; page-preservation and independent inspection receipts accompany publication
Canonical progress versus boundary promotion	Diagnostic progress entry records this run; no boundary-ledger promotion, calibrated S or independent-host claim

Independent C++ arithmetic regenerates the fixtures, replays every completed scalar and verifies every exported IC decomposition witness. File and execution-matrix audits bind source, binaries, inputs and records. The post-hoc audit is metered separately and is not silently assigned as a per-target production-verification charge.

Measured sources were frozen at c6e8daafb5a59d27e1be88a2f61f0bd6ee83ef03 and prospective binary/matrix metadata at ba94c10811ab48da60c9af1824c1eb767a9e17ec. The final independent V3 replay records 1,237 hash bindings, 239 distinct bound files and 336 file snapshots. The separate collect-only extractor rechecks all 336 snapshot hashes and exports 32 process rows, 192 cohort query rows and 26 auxiliary target rows. Source pins cover selected relevant sources and Cargo.lock, not a full compiled dependency-closure proof.

Preserved failure: the frozen V2 checker reported invalid_rows=227 after metadata-binding failures. Native diagnosis found 127 copied floating-point tokens, each exactly one decoded f64 ULP from its original after Rust parsing/reserialization. V3 changes only four observation-copy comparisons, allowing adjacent finite f64 values at explicit diagnostic paths. Integer/coordinate/scalar/seed/counter/nanosecond/hash/configuration gates remain exact; two ULPs and one-nanosecond changes are rejected. The original frozen collect command then failed exact observation reconstruction; a separate versioned collect-only program uses arbitrary-precision JSON tokens and the V3 file seal, without recalculating clocks or launching solvers. Original failures, sources, observations and all measured bytes remain unchanged. This is a post-hoc validation/extraction correction, not a newly measured solver improvement.

Sources: Final independent replay, V3 numeric-copy correction, Separate collect-only extraction, Pre-timing checks.

SECTION 15

What changes in the measurement approach

For the measured retained-base one-target workload and the actual 32-target shared workload, the optimized rho implementation has lower complete-process cost. IC's secondary producer-query mean is lower, but its target-block and crossed intervals include equal cost, its upper query tail is higher, and its clock omits validation/input/output work included by rho. The observed data therefore support using this rho configuration for these two executed workload sizes on this host. They do not establish global optimality or an executed larger-batch IC crossover.

Measurement change	Why it matters / next validation
Align target-dependent clocks before an online headline	Include matching input/Q validation, output and independent-check boundaries; keep present producer query ratios secondary
Add independent preparation clusters and hosts	Three shared preparations on one guest give limited interval resolution; target/seed repeats do not replace setup/host replication
Distribute A/A controls across timing epochs	Current five pairs precede the multi-hour matrix; their initial range cannot bound later host/frequency drift
Retain all cost regimes and distributions	Keep actual cold and genuine shared totals, means, conventional medians, slow cases, failures, memory and internal restart work
Price original K600 selection/construction	This unavailable additive cost remains open for a curve-only cold-start or complete deployment crossover
Fix producer JSON round trips prospectively	V3 verifier and token-preserving collect-only corrections are post-hoc audit/extraction changes; a future producer fix needs a new frozen protocol and new measurements
Execute any claimed larger crossover	N>32 remains modeled; charge fresh rho seed-specific jumps, retain nonfinite uncertainty, and do not infer globally optimal amortized rho
Keep historical and current panels separate	No multiplication of historical gains or pairing old IC timings with this newly optimized rho

The broad curve/vector-basis synthesis and all earlier degree-83 studies follow as the unchanged 70-page historical appendix. New figures are study-local diagnostics. Canonical admitted boundary graphs and ledger values change only when their own claim contracts are met.

The frozen protocol, create-new execution matrix, exact build commands, source/binary/input hashes, raw records, independent replay, versioned post-hoc corrections, native statistics and supplement are committed with this report. Reproduce into new directories. Benchmark PDFs/SVG/150 dpi PNGs are browsing copies; study records remain authoritative. The renderers only format native values and inspect layout; they do not recompute results. Native evidence is pinned at fb26ccaef5c8f5e97aa81b3503d42e5501b030e4. Authoritative evidence.

Sources: Executed workload statistics, Accounting protocol, Preserved numeric-copy diagnosis.